



Trabajo Práctico Comunicaciones IPv6 2026

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Objetivo 1 - Wireshark

```
alumno@pelle001:~$ ifconfig
br-7a2b853679af: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.18.0.1 netmask 255.255.0.0 broadcast 172.18.255.255
    ether aa:52:35:fc:36:d2 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 92:39:91:1f:ea:e5 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 30 overruns 0 carrier 0 collisions 0

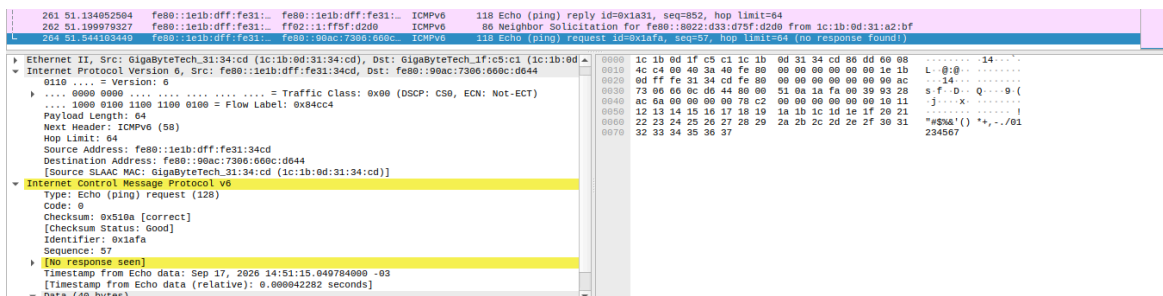
enp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.144.124 netmask 255.255.255.0 broadcast 192.168.144.255
    inet6 fe80::1e1b:dff:fe31:34cd prefixlen 64 scopeid 0x20<link>
    ether 1c:1b:0d:31:34:cd txqueuelen 1000 (Ethernet)
    RX packets 24042 bytes 23169283 (23.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8671 bytes 1383838 (1.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Bucle local)
    RX packets 440 bytes 42622 (42.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 440 bytes 42622 (42.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

ifconfig (*ipconfig* para Linux), analizando las direcciones asignadas.

Podemos observar diferentes interfaces: *docker0*, *enp1s0* y *lo*, donde las que poseen direcciones IPv6 son *enp1s0* `fe80::1e1b:dff:fe31:34cd/64` perteneciendo a un tipo de ip Link-Local y *lo* `::1` haciendo referencia al loopback.

Realizamos ping sobre nuestro propio equipo (loopback) para verificar nuestra conectividad y no obtuvimos problema. Luego, realizamos ping a otro equipo del mismo enlace. Observamos en Wireshark el resultado del ping:

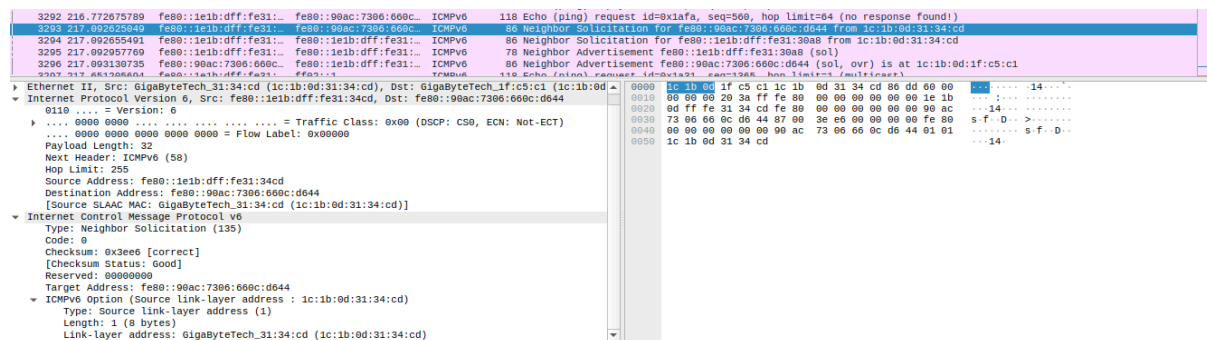


Se pueden observar los campos de la cabecera IPv6.

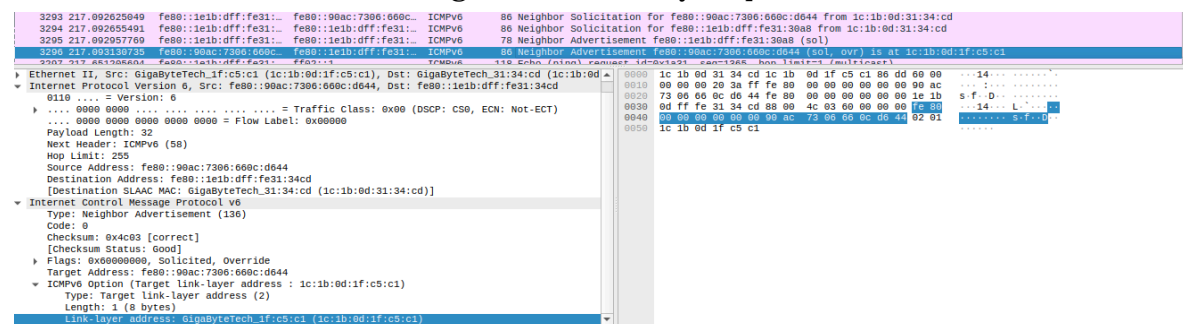
- Version (identifica la versión del protocolo): 6.
- Traffic Class (identifica y diferencia paquete por clases y prioridad): 0x00.
- Flow Label (identifica y diferencia paquetes del mismo flujo): 0x84cc4.
- Payload Length (tamaño en bytes de los datos enviados): 64.
- Next Header (encabezado que sigue al encabezado IPv6): ICMPv6.
- Hop Limit (número máximo de routers que el paquete puede pasar antes de ser descartado): 64.
- Source Address (dirección de origen del paquete): fe80::1e1b:dff:fe31:34cd.
- Destination Address (dirección de destino del paquete): fe80::90ac:7406:660c:d644.

Notamos que en el Header ICMPv6 el tipo de mensaje que se envió fue un *Echo Request*. A su vez, se obtuvo, *no response found*, es decir, no se encontró una respuesta asociada a esta petición.

También, pudimos identificar los procesos de Neighbor Discovery realizados a la misma PC que le realizamos el *ping*.



Neighbor Discovery request.



Neighbor Discovery response.

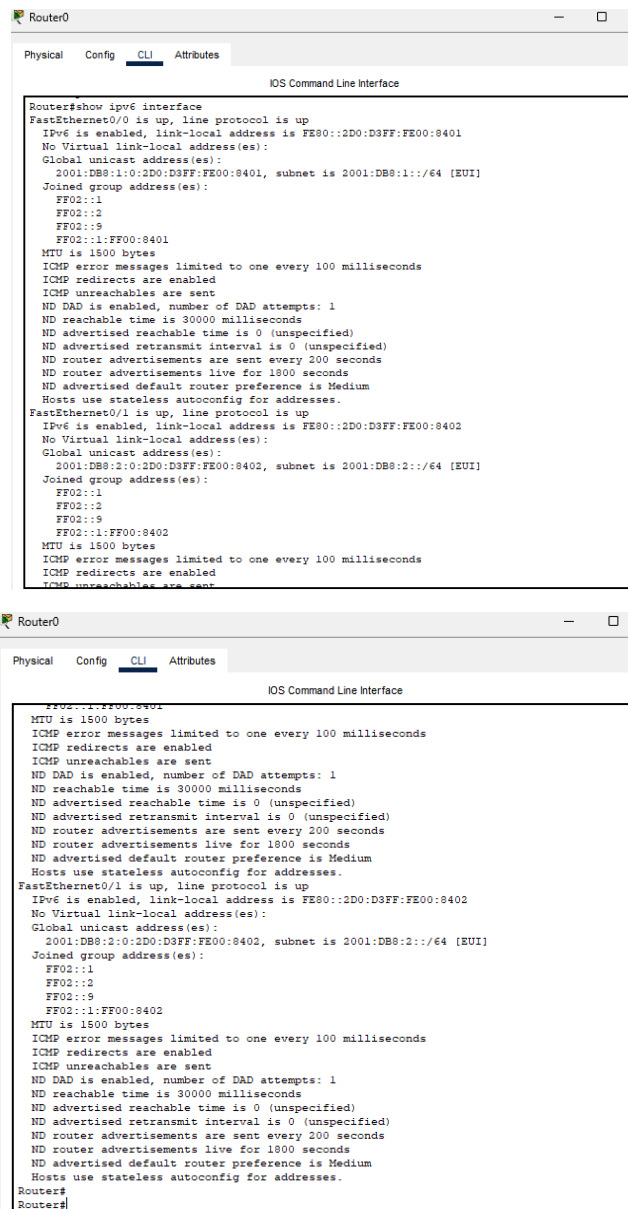
El procedimiento consiste en lo siguiente: el equipo que solicita el descubrimiento pregunta por la dirección fe80::90ac:7406:660c:d644 e informa

la suya. El vecino responde que esa dirección corresponde a la MAC 1c:1b:0d:1f:c5:c1.

Objetivo 2 - Cisco Packet Tracer

Ejecutamos sobre las interfaces de cada Router el comando `Router# show ipv6 interface` y obtuvimos lo siguiente:

- Router0



The image shows two screenshots of the Cisco Packet Tracer CLI for Router0. The first screenshot shows the output of the command `Router#show ipv6 interface`, which displays the configuration for two interfaces: FastEthernet0/0 and FastEthernet0/1. Both interfaces are configured with IPv6, including link-local addresses, global unicast addresses, and joined group addresses. The second screenshot shows the output of the command `Router#`, which displays the configuration for the FastEthernet0/1 interface, including link-local addresses, global unicast addresses, and joined group addresses.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#show ipv6 interface
FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::2D0:D3FF:FE00:8401
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:1:0:2D0:D3FF:FE00:8401, subnet is 2001:DB8:1::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FE00:8401
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
FastEthernet0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::2D0:D3FF:FE00:8402
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:2:0:2D0:D3FF:FE00:8402, subnet is 2001:DB8:2::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FE00:8402
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
Router#
Router#
```

Interfaz Router 0, se puede observar su correcta configuración para soportar IPv6.

- Router1:

The image shows two screenshots of the Router1 CLI interface. The top screenshot shows the output of the command 'show ipv6 interface' for FastEthernet0/0 and FastEthernet0/1. The bottom screenshot shows the output of the command 'show ipv6 interface' for FastEthernet0/0 and FastEthernet0/1, with the output for FastEthernet0/0 repeated.

```
Router1#show ipv6 interface
FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::20A:F3FF:FEE0:9C01
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:3:0:20A:F3FF:FEE0:9C01, subnet is 2001:DB8:3::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FFE0:9C01
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
FastEthernet0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::20A:F3FF:FEE0:9C02
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:2:0:20A:F3FF:FEE0:9C02, subnet is 2001:DB8:2::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FFE0:9C02
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
```

```
Router1#show ipv6 interface
FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::20A:F3FF:FEE0:9C01
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:3:0:20A:F3FF:FEE0:9C01, subnet is 2001:DB8:3::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FFE0:9C01
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
FastEthernet0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::20A:F3FF:FEE0:9C02
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:2:0:20A:F3FF:FEE0:9C02, subnet is 2001:DB8:2::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::9
  FF02::1:FFE0:9C02
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
Router1#
```

Interfaz Router 1, se puede observar su correcta configuración para soportar IPv6.

Con los resultados obtenidos arriba, completamos la siguiente tabla:

Dispositivo	IPv6 habilitado	Dirección local (link-local)	Dirección global
Router0 Fa0/0	Sí	FE80::2D0:D3FF:FE00:8401	2001:DB8:1:0:2D0:D3FF:FE00:8401
Router0 Fa0/1	Sí	FE80::2D0:D3FF:FE00:8402	2001:DB8:2:0:2D0:D3FF:FE00:8402
Router1 Fa0/0	Sí	FE80::20A:F3FF:FEE0:9C01	2001:DB8:3:0:20A:F3FF:FEE0:9C01
Router1 Fa0/1	Sí	FE80::20A:F3FF:FEE0:9C02	2001:DB8:2:0:20A:F3FF:FEE0:9C02

Las direcciones IPv6 constan de 128 bits a diferencia de los 32 bits de IPv4. La interfaz Fa0/0 del Router0 tiene como prefijo de subred 2001:db8:1::/64, mientras que el ID de la interfaz es: 2d0:d3ff:fe00:8401. La dirección MAC de la interfaz es 00D0.D300.8401. Actualmente no existe relación entre MAC e IPv6.

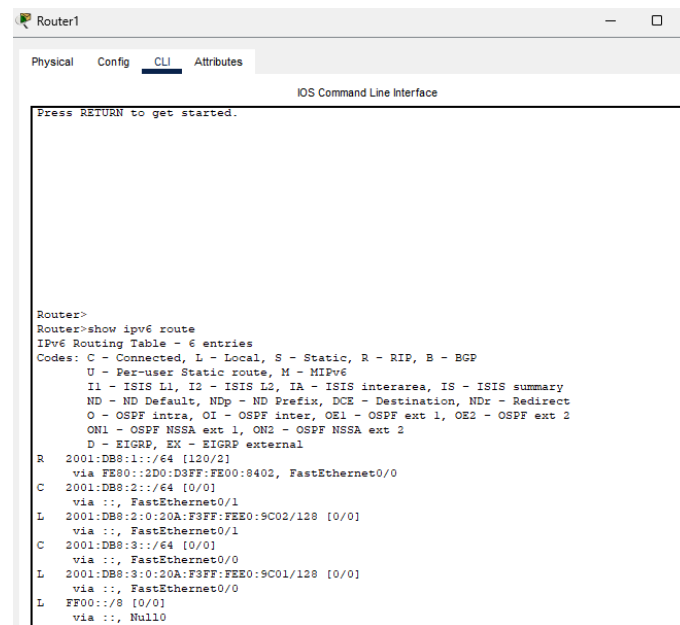
Con Router# show ipv6 route obtuvimos la tabla de ruteo para cada router:

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#show ipv6 route
^
% Invalid input detected at '^' marker.
Router(config-if)# show ipv6 route
^
% Invalid input detected at '^' marker.
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ipv6 route
IPv6 Routing Table - 6 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C 2001:DB8:1::/64 [0/0]
  via ::, FastEthernet0/0
L 2001:DB8:1:0:2D0:D3FF:FE00:8401/128 [0/0]
  via ::, FastEthernet0/0
C 2001:DB8:2::/64 [0/0]
  via ::, FastEthernet0/1
L 2001:DB8:2:0:2D0:D3FF:FE00:8402/128 [0/0]
  via ::, FastEthernet0/1
R 2001:DB8:3::/64 [120/2]
  via FE80::20A:F3FF:FEE0:9C01, FastEthernet0/1
L FF00::/8 [0/0]
  via ::, Null0

```

Tabla de ruteo de Router 0.



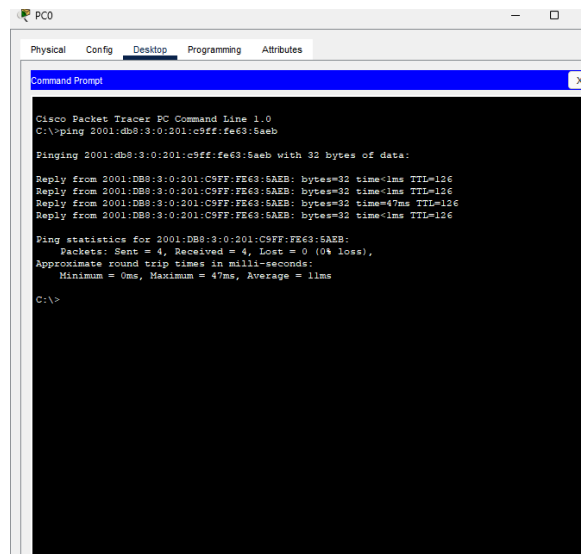
```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>
Router>show ipv6 route
IPv6 Routing Table - 6 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
Codes: U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
R 2001:DB8:1::/64 [120/2]
   via FE80::2D0:D3FF:FE00:8402, FastEthernet0/0
C 2001:DB8:2::/64 [0/0]
   via ::, FastEthernet0/1
L 2001:DB8:2:0:20A:F3FF:FE00:9C02/128 [0/0]
   via ::, FastEthernet0/1
C 2001:DB8:3::/64 [0/0]
   via ::, FastEthernet0/0
L 2001:DB8:3:0:20A:F3FF:FE00:9C01/128 [0/0]
   via ::, FastEthernet0/0
L FF00::/8 [0/0]
   via ::, Null0
```

Tabla de ruteo de Router 1.

Observamos que se envían 4 paquetes y se reciben 4 por lo que confirmamos que el tráfico cruza los dos routers.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:3:0:201:c9ff:fe63:5aeb

Pinging 2001:db8:3:0:201:c9ff:fe63:5aeb with 32 bytes of data:

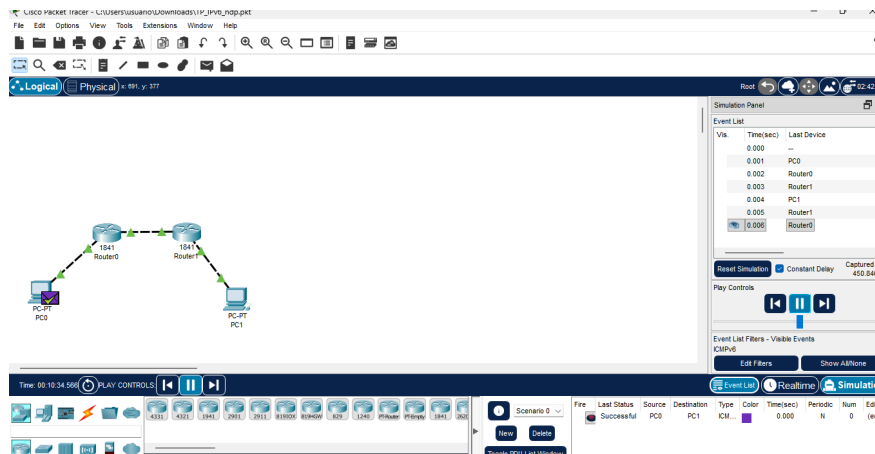
Reply from 2001:DB8:3:0:201:C9FF:FE63:5AEB: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:3:0:201:C9FF:FE63:5AEB: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:3:0:201:C9FF:FE63:5AEB: bytes=32 time<4ms TTL=126
Reply from 2001:DB8:3:0:201:C9FF:FE63:5AEB: bytes=32 time<1ms TTL=126

Ping statistics for 2001:DB8:3:0:201:C9FF:FE63:5AEB:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 47ms, Average = 11ms

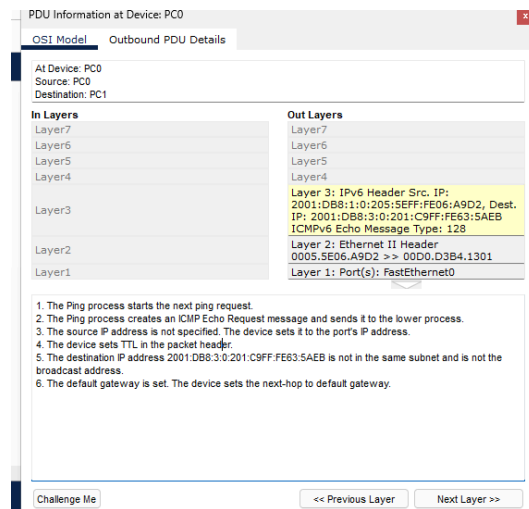
C:\>
```

Verificación conectividad extremo a extremo utilizando ping de PC0 a PC1.

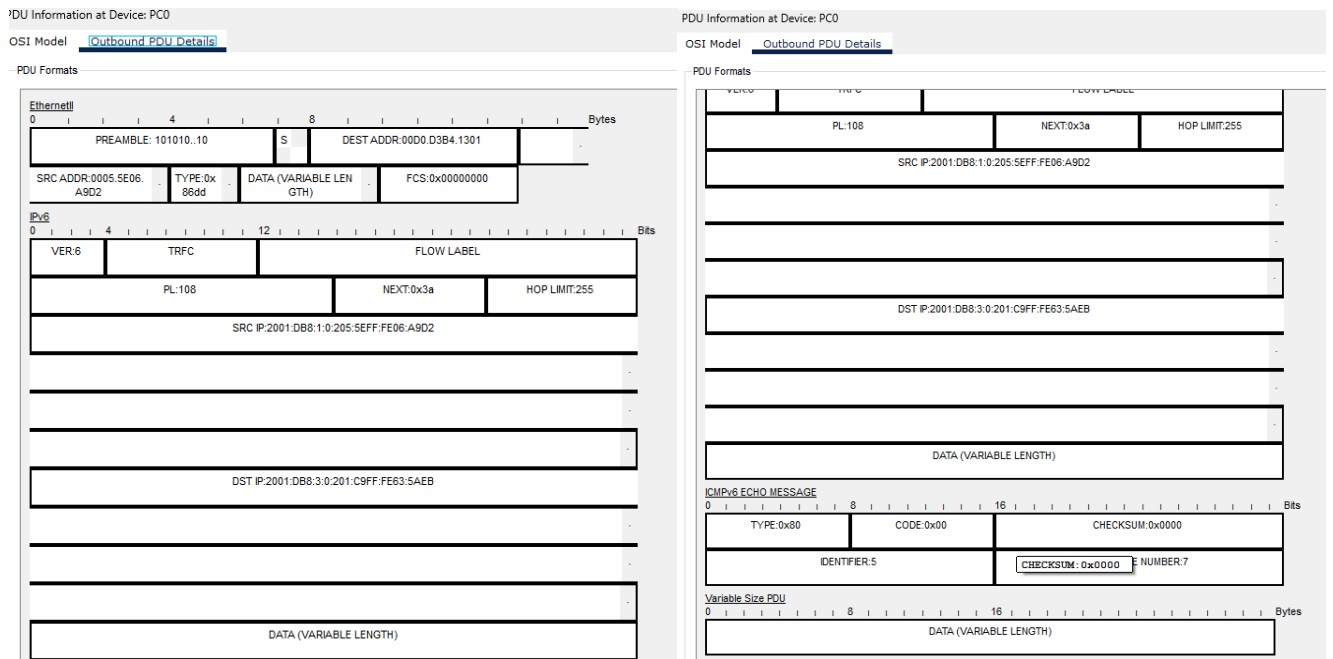
Comenzamos con la simulación del envío de paquetes desde la PC0 hacia la PC1



Simulación de flujo de datos de PC0 a PC1.



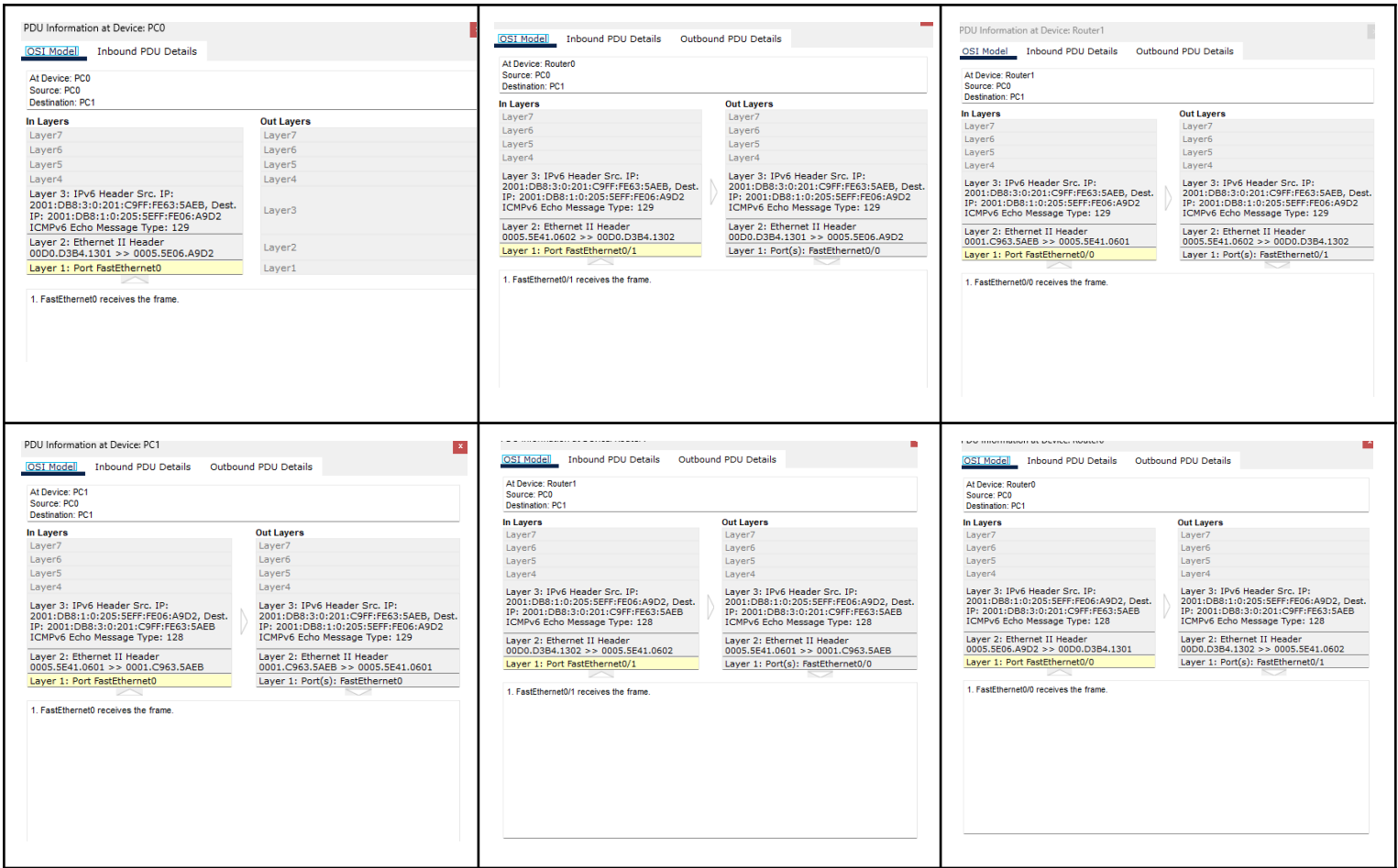
Información del paquete y encabezado. Muestra del modelo OSI del mensaje.



Detalles de la PDU de salida.

En resumen, los detalles obtenidos son:

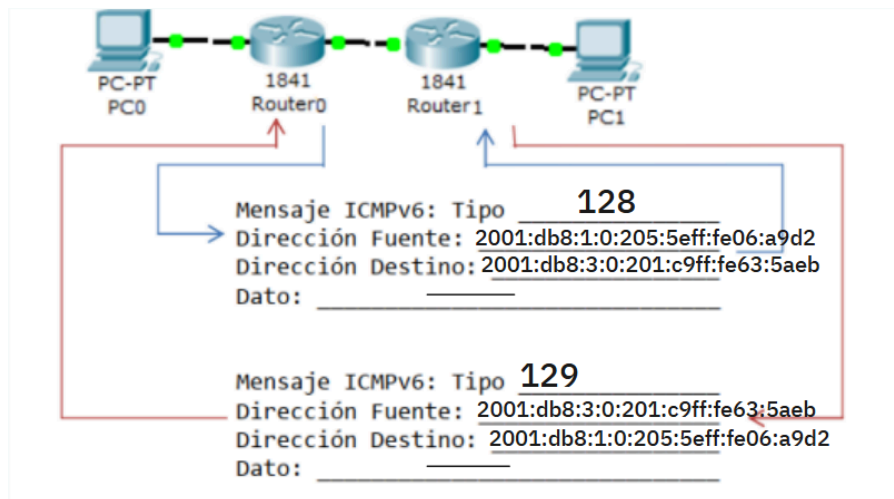
Cabecera	Campo	Valor
Ethernet II	Destino / Origen / Type	00D0 . D3B4 . 1301 (Router0) / 0005 . 5E06 . A9D2 (PC0) / 0x86dd (IPv6)
IPv6	Version / Next / Hop Limit	6 / 0x3a = 58 (ICMPv6) / 255
IPv6	Payload Length	108
IPv6	Origen / Destino	2001:DB8:1:0:205:5EFF:FE06:A9D2 / 2001:DB8:3:0:201:C9FF:FE63:5AEB
ICMPv6 Echo	Type / Code	0x80 = 128 (Echo Request) / 0x00
ICMPv6 Echo	Identifier / Sequence	5 / 7
ICMPv6 Echo	Checksum	0x0000 (el simulador no lo calcula)



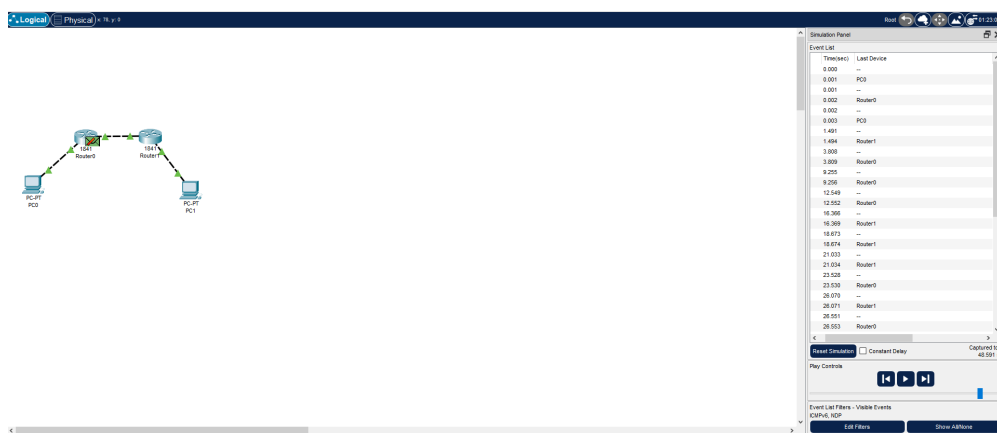
Visualización de la secuencia de Mensajes, de ida y vuelta.

Dispositivo	Puerto	Tipo	Trama (MAC origen >> destino)
PC0 (salida)	Fa0	128 (Request)	0005.5E06.A9D2 >> 00D0.D3B4.1301
Router0 (entrada)	Fa0/0	128	0005.5E06.A9D2 >> 00D0.D3B4.1301
Router0 (salida)	Fa0/1	128	00D0.D3B4.1302 >> 0005.5E41.0602
Router1 (entrada)	Fa0/1	128	00D0.D3B4.1302 >> 0005.5E41.0602
Router1 (salida)	Fa0/0	128	0005.5E41.0601 >> 0001.C963.5AEB
PC1 (entrada)	Fa0	128	0005.5E41.0601 >> 0001.C963.5AEB
PC1 (salida)	Fa0	129 (Reply)	0001.C963.5AEB >> 0005.5E41.0601
Router1 (entrada)	Fa0/0	129	0001.C963.5AEB >> 0005.5E41.0601
Router1 (salida)	Fa0/1	129	0005.5E41.0602 >> 00D0.D3B4.1302
Router0 (entrada)	Fa0/1	129	0005.5E41.0602 >> 00D0.D3B4.1302
Router0 (salida)	Fa0/0	129	00D0.D3B4.1301 >> 0005.5E06.A9D2
PC0 (entrada)	Fa0	129	00D0.D3B4.1301 >> 0005.5E06.A9D2

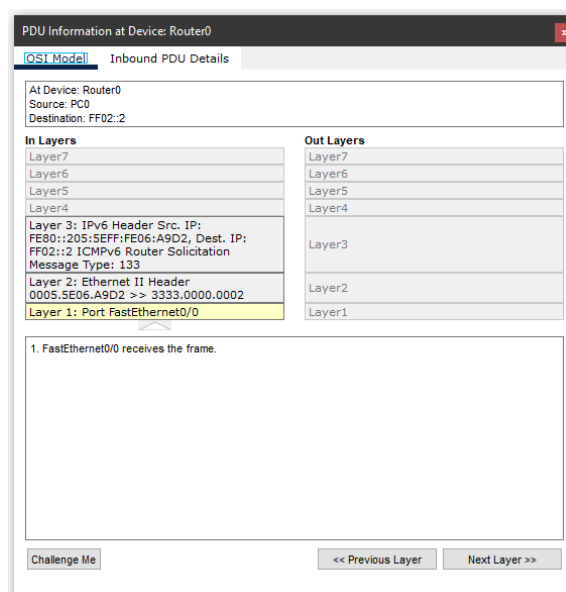
Tabla de referencia para la imagen anterior.



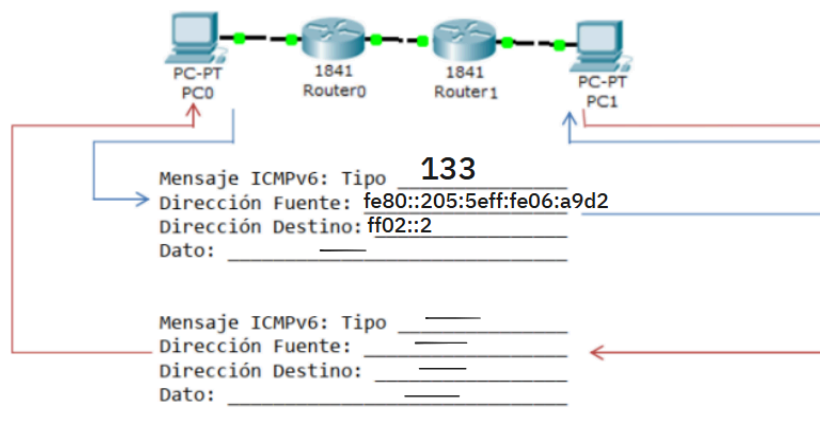
Tarea 4, punto d.



Fallo en descubrimiento al hacer Auto Config en IPv6.



Solicitud de PC0 al Router 0 mediante un Router Solicitation.



La respuesta esperada es Router Advertisement (tipo 134) de Router 0 pero no pudimos obtener el resultado.

(Tarea 4, punto f).

Conclusión

En el trabajo utilizamos dos programas que nos permitieron analizar el funcionamiento de IPv6 tanto en un equipo real con rastreo de paquetes (Wireshark) como en una red simulada (Cisco Packet Tracer), y pudimos comprobar y poner en práctica varios conceptos vistos en teoría. Algunos de estos conceptos analizados fueron el contenido de mensajes ICMPv6 que se intercambian entre anfitriones, así como el funcionamiento de procesos como Neighbor Discovery, de manera detallada.

En cuanto al desarrollo del trabajo práctico en sí, fue algo complicado armar el documento de informe debido a la falta de detalle sobre qué cosas específicamente escribir. Aún así, por la parte práctica fue útil ver en tiempo real cosas dadas en la teoría, de manera que se afianzaron los conocimientos teóricos vistos en clase.